Labsheet-5

```
import pandas as pd
import numpy as np
import plotly.express as px
import ipywidgets as widgets
from IPython.display import display, clear_output
```

```
np.random.seed(42)

regions = ['East', 'West', 'Central', 'South']
categories = ['Furniture', 'Office Supplies', 'Technology']
sub_categories = {
    'Furniture': ['Chairs', 'Tables', 'Bookcases'],
    'Office Supplies': ['Binders', 'Paper', 'Storage'],
    'Technology': ['Phones', 'Machines', 'Accessories']
}
segments = ['Consumer', 'Corporate', 'Home Office']
region_states = {
    'East': ['NY', 'PA', 'NJ', 'MA'],
    'West': ['CA', 'WA', 'OR', 'NV'],
    'Central': ['TX', 'IL', 'OH', 'MI'],
    'South': ['FL', 'GA', 'NC', 'TN']
}

n = 1000
dates = pd.date_range('2023-01-01', '2024-12-31', periods=n)

rows = []
for i in range(n):
    region = np.random.choice(regions)
    category = np.random.choice(categories)
    sub_category = np.random.choice(sub_categories[category])
    segment = np.random.choice(segments)
    state = np.random.choice(region_states[region])
    sales = round(float(np.random.uniform(20, 3000)), 2)
    profit = round(sales * float(np.random.uniform(-0.2, 0.35)), 2)
    quantity = int(np.random.randint(1, 10))
    rows.append([dates[i], region, state, category, sub_category, segment, sales, profit,

df = pd.DataFrame(rows, columns=['Order Date', 'Region', 'State', 'Category',
                                 'Sub-Category', 'Segment', 'Sales', 'Profit', 'Quantity
df['Order Date'] = pd.to_datetime(df['Order Date'])
df['Month'] = df['Order Date'].dt.to_period('M').astype(str)

df.head()
```

	Order Date	Region	State	Category	Sub-Category	Segment	Sales	Profit	Qu
0	2023-01-01 00:00:00.000000000	Central	MI	Furniture	Bookcases	Home Office	1804.00	-206.00	
1	2023-01-01 17:32:15.135135135	Central	MI	Technology	Accessories	Consumer	445.74	70.42	
2	2023-01-02 11:04:30.270270270	West	NV	Office Supplies	Paper	Corporate	2976.79	415.60	
3	2023-01-03 04:36:45.405405405	East	NJ	Furniture	Chairs	Home Office	1211.59	-211.22	
4	2023-01-03 22:09:00.540540540	South	FL	Technology	Phones	Home Office	1830.48	-194.42	

Next steps: Generate code with df | New interactive sheet

```
region_filter = widgets.SelectMultiple(options=regions, value=regions, description='Reg
category_filter = widgets.SelectMultiple(options=categories, value=categories, descript
segment_filter = widgets.SelectMultiple(options=segments, value=segments, description='

output = widgets.Output()

def update_dashboard(change=None):
    with output:
        clear_output(wait=True)

        filtered = df[
            df['Region'].isin(region_filter.value) &
            df['Category'].isin(category_filter.value) &
            df['Segment'].isin(segment_filter.value)
        ]

        if filtered.empty:
            print("No data for this filter combination.")
            return

        total_sales = filtered['Sales'].sum()
        total_profit = filtered['Profit'].sum()
        total_orders = len(filtered)
        avg_order_value = total_sales / total_orders

        print("KEY PERFORMANCE INDICATORS")
        print(f"  Total Sales          : ${total_sales:,.2f}")
        print(f"  Total Profit         : ${total_profit:,.2f}")
        print(f"  Number of Orders     : {total_orders:,}")
        print(f"  Average Order Value  : ${avg_order_value:,.2f}")
        print("-" * 50)

        fig1 = px.bar(filtered.groupby('Category', as_index=False)['Sales'].sum(),
                      x='Category', y='Sales', color='Category', title='Sales by Catego
        fig1.show()

        monthly = filtered.groupby('Month', as_index=False)['Sales'].sum()
        fig2 = px.line(monthly, x='Month', y='Sales', markers=True, title='Monthly Sale
```

```
        fig2.show()

        fig3 = px.pie(filtered, values='Sales', names='Segment', title='Sales Share by
        fig3.show()

        fig4 = px.treemap(filtered, path=['Category', 'Sub-Category'], values='Sales',
                          title='Sales Treemap (click a box to drill down)')
        fig4.show()

        state_sales = filtered.groupby('State', as_index=False)['Sales'].sum()
        fig5 = px.choropleth(state_sales, locations='State', locationmode='USA-states',
                             color='Sales', scope='usa', title='Sales by State')
        fig5.show()

region_filter.observe(update_dashboard, names='value')
category_filter.observe(update_dashboard, names='value')
segment_filter.observe(update_dashboard, names='value')
```

```
display(widgets.HBox([region_filter, category_filter, segment_filter]))
display(output)
update_dashboard()
```

Start coding or generate with AI.

South

```
KEY PERFORMANCE INDICATORS
  Total Sales          : $39,679.59
  Total Profit         : $3,187.22
  Number of Orders     : 22
  Average Order Value  : $1,803.62
--------------------------------------------------
```